

MEL1010 Engineering Mechanics

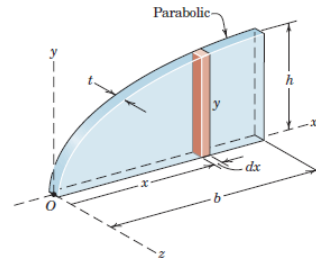
Tutorial 10 Dynamics using Work-Energy, Impulse-Momentum Approach and Rigid body Dynamics in 3 Dimensions

30 Marks

Answer all of the following questions. Mention valid assumptions wherever applicable

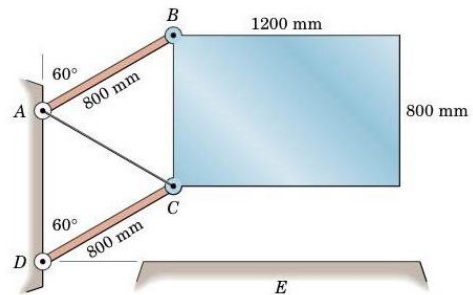
- 1 The upper edge of the thin homogeneous plate of mass m is parabolic with a vertical slope at the origin O . Determine its mass moments of inertia about the x -, y -, and z -axes.

$$\text{Ans. } I_{xx} = \frac{1}{5}mh^2, I_{yy} = \frac{3}{7}mb^2, I_{zz} = m\left(\frac{h^2}{5} + \frac{3b^2}{7}\right)$$



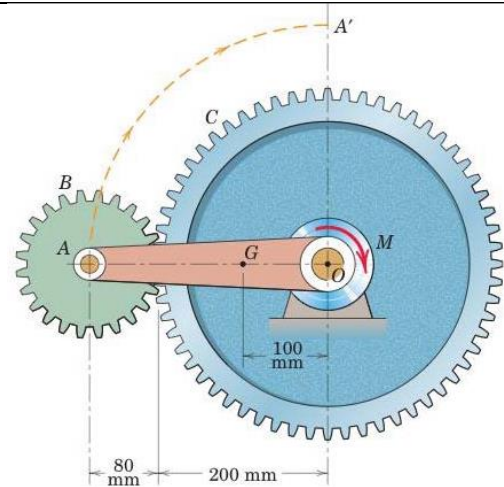
- 2 The uniform rectangular plate has a mass of 300 kg and is supported in the vertical plane by the two parallel links of negligible mass and by the cable AC . If the cable suddenly breaks, determine the angular velocity ω of the links an instant before the plate strikes the horizontal surface E . Also find the force in member DC at the same instant. (Use Work Energy Method)

$$\text{Ans. } \omega = 3 \frac{\text{rad}}{\text{s}}, F_{DC} = 1472 \text{ N}$$



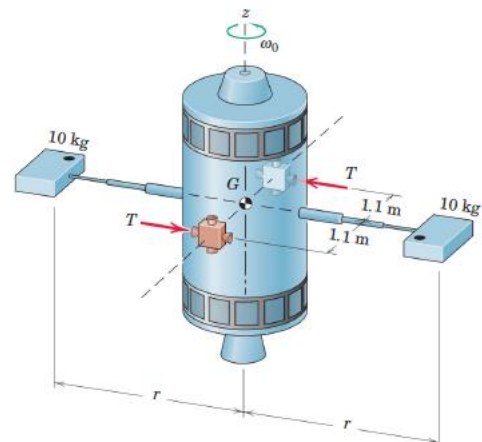
- 3 For the assembly shown, arm OA has a mass of 0.8 kg and a radius of gyration about O of 140 mm. Gear B has a mass of 0.9 kg and may be treated as a solid circular disk. Gear C is fixed in the vertical plane and cannot rotate. If a constant moment $M = 4 \text{ N}\cdot\text{m}$ is applied to arm OA , initially at rest in the horizontal position shown, calculate the velocity v of point A as it reaches the top at A' . (Use Work Energy Method)

$$\text{Ans. } v = 1.976 \text{ m/s}$$



- 4 Two small variable-thrust jets are actuated to keep the spacecraft angular velocity about the z -axis constant at $\omega_0 = 1.25 \text{ rad/s}$ as the two telescoping booms are extended from $r_1 = 1.2 \text{ m}$ to $r_2 = 4.5 \text{ m}$ at a constant rate over a 2-min period. Determine the necessary thrust T for each jet as a function of time where $t = 0$ is the time when the telescoping action is begun. The small 10 kg experiment modules at the ends of the booms may be treated as particles, and the mass of the rigid booms is negligible. (Use Impulse Momentum Method)

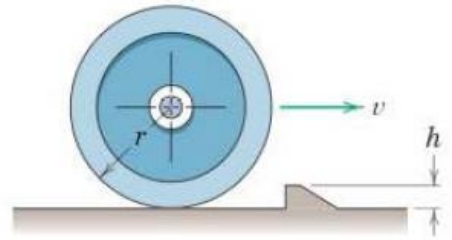
$$\text{Ans. } T = 0.750 + 0.01719t$$



- 5 Determine the minimum velocity v which the wheel must have to just roll over the obstruction. The centroidal radius of gyration of the wheel is k , and it is assumed that the wheel does not slip.

(Use Impulse Momentum Method)

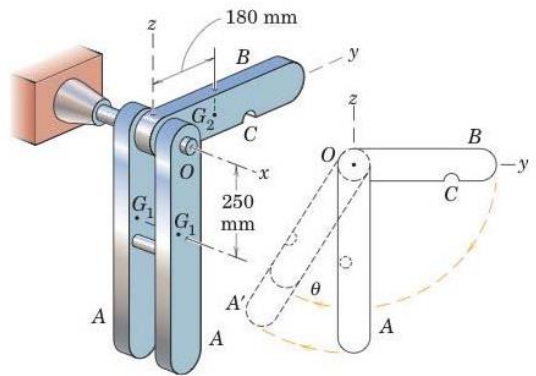
Ans. $v = \frac{r}{k^2 + r^2 - rh} \sqrt{2gh(k^2 + r^2)}$



- 6 The three bars are free to rotate about the fixed horizontal x -axis at O . Each of the bars A has a mass of 20 kg , a radius of gyration about the x -axis of 300 mm , and a mass center at G_1 . The 8 kg bar B , with a radius of gyration about the same axis of 220 mm and a mass center at G_2 , is released from rest in the horizontal (y -axis) position and becomes attached to bars A by a latch at C at the bottom of its swing. Calculate the angle θ through which the three bars rotate as a unit and find the loss $|\Delta E|$ of energy due to the impact.

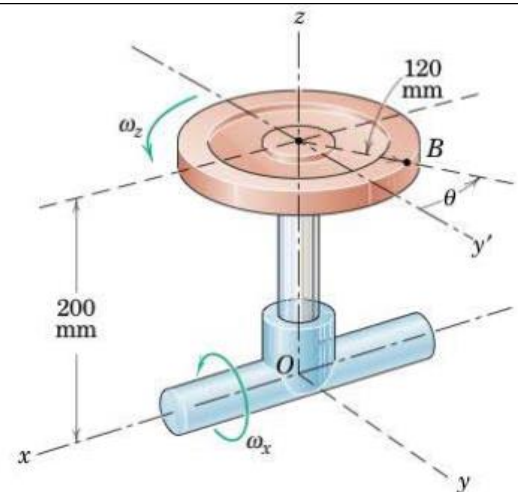
(Use Impulse Momentum Method)

Ans. $\theta = 8.97^\circ$, $|\Delta E| = 12.75\text{ J}$



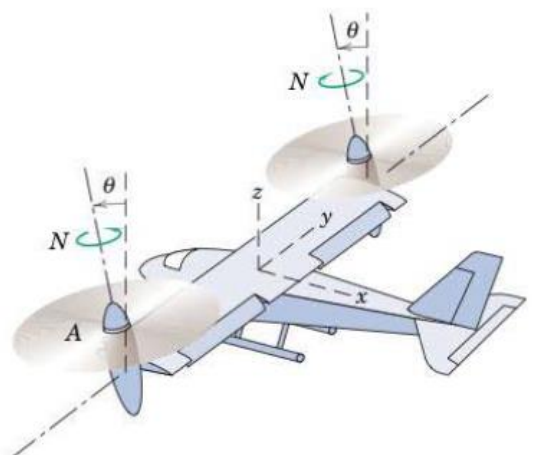
- 7 The circular disk of 120 mm radius rotates about the z -axis at the constant rate $\omega_z = 20\text{ rad/s}$, and the entire assembly rotates about the fixed x -axis at the constant rate $\omega_x = 10\text{ rad/s}$. Calculate the magnitudes of the velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of point B for the instant when $\theta = 30^\circ$.

Ans. $v = 3.95\text{ m/s}$, $a = 72.2\text{ m/s}^2$

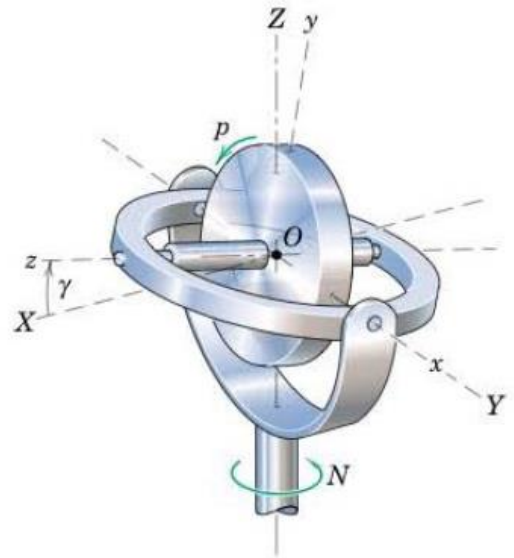


- 8 An unmanned radar-radio controlled aircraft with tilt-rotor propulsion is being designed for reconnaissance purposes. Vertical rise begins with $\theta = 0$ and is followed by horizontal flight as θ approaches 90° . If the rotors turn at a constant speed N of 360 rev/min , determine the angular acceleration α of rotor A for $\theta = 30^\circ$ if $\dot{\theta}$ is constant at 0.2 rad/s .

Ans: $\alpha_x = -1.2\pi(\sqrt{3}\mathbf{i} + \mathbf{k})\text{ rad/s}^2$



- 9 The gyro rotor is spinning at the constant rate $p = 100 \text{ rev/min}$ relative to the x - y - z axes in the direction indicated. If the angle γ between the gimbal ring and horizontal X - Y plane is made to increase at the rate of 4 rad/sec and if the unit is forced to precess about the vertical at the constant rate $N = 20 \text{ rev/min}$, calculate the angular momentum $\mathbf{H}_O$ of the rotor when $\gamma = 30^\circ$. The axial and transverse moments of inertia are $I_{zz} = 5(10^{-3}) \text{ lb-ft-sec}^2$ and $I_{xx} = I_{yy} = 2.5(10^{-3}) \text{ lb-ft-sec}^2$
- Ans: $\mathbf{H}_O = -0.01 \mathbf{i} + 0.0045 \mathbf{j} + 0.0576 \mathbf{k} \text{ lb-ft-sec}$



- 10 The solid circular disk of mass $m = 2 \text{ kg}$ and radius $r = 100 \text{ mm}$ rolls in a circle of radius $b = 200 \text{ mm}$ on the horizontal plane without slipping. If the center-line OC of the axle of the wheel rotates about the z -axis with an angular velocity $\omega = 4\pi \text{ rad/s}$, determine the expression for the angular momentum of the disk with respect to the fixed-point O . Also compute the kinetic energy of the wheel.
- Ans: $\mathbf{v}_A = 0.251(-\mathbf{j} + 4.25\mathbf{k}) \text{ N.m.s}$
 $T = 9.87 \text{ J}$

